Last name _____

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LARSON—MATH 310–CLASSROOM WORKSHEET 17
Matrix powers, QR factorization and Inverses.

Review

1. For matrices A, B, $(AB)^T = B^T A^T$.
2. Matrices A and B define linear functions. What linear function does AB define?

Matrix Powers (Sec. 10.3)
Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

1. Find $A \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

2. Find A^2.

3. Find $A^2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

4. Find A^3.

5. Find $A^3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

QR Factorization (Sec. 10.4)

6. Let $\vec{a}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \vec{a}_2 = \begin{bmatrix}0\\0\\1\end{bmatrix}, \vec{a}_3 = \begin{bmatrix}1\\1\\1\end{bmatrix}$ Use the **Gram-Schmidt algorithm** to find orthonormal $\vec{q}_1$, $\vec{q}_2$, $\vec{q}_3$. Write each $\vec{a}_i$ in terms of $\vec{q}_i$'s.

7. Write $A = [\vec{a}_1\vec{a}_2\vec{a}_3]$ as $A = [\vec{q}_1\vec{q}_2\vec{q}_3]R$, where

$$\vec{a}_1 = R_{1,1}\vec{q}_1 + R_{2,1}\vec{q}_2 + R_{3,1}\vec{q}_3.$$

$$\vec{a}_2 = R_{1,2}\vec{q}_1 + R_{2,2}\vec{q}_2 + R_{3,2}\vec{q}_3.$$

$$\vec{a}_3 = R_{1,3}\vec{q}_1 + R_{2,3}\vec{q}_2 + R_{3,3}\vec{q}_3.$$

Let

$$A = \begin{bmatrix}0 & 1 & 2 & 1 & 0 & 1\\ 1 & 0 & 0 & 1 & 0 & 3\\ 0 & 0 & 0 & 0 & 0 & 0\end{bmatrix}$$

8. Let the columns of A be $\vec{a}_1$,..., $veca_6$. Use the **Gram-Schmidt algorithm** to find orthonormal $\vec{q}_1, \ldots, \vec{q}_r$ (where r is the number of produced $\vec{q}$ vectors). Write each $\vec{a}_i$ in terms of $\vec{q}_i$'s. Note that is a column $\vec{a}_i$ is not linearly independent from the previously produced $\vec{q}$'s, it can be written in terms of these.

9. Write $A = [\vec{a}_1 \ldots \vec{a}_6]$ as $A = [\vec{q}_1 \ldots \vec{q}_r]R$, where R *codes* each vector $\vec{a}_i$ in terms of the previously produced $\vec{q}_j$'s. Get $A = QR$. Check.